

Student Guide — primary / module-06- moon-phases-tithi

IKS Curriculum

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Student Workbook

Student Workbook — Moon Phases and Tithi (Primary)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *candra* — _____
- *pakṣa* — _____
- *śukla pakṣa* — _____
- *kṛṣṇa pakṣa* — _____
- *tithi* — _____
- *amāvasyā* — _____
- *pūrṇimā* — _____
- *pañcāṅga* — _____

Day-by-day notes

Day 1 — The Moon and Its Shapes

What I learned:

One question I have:

Day 2 — Two Fortnights — Pakṣa

What I learned:

One question I have:

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What I learned:

One question I have:

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One question I have:

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Day 8 — Modern Lunar Science

What I learned:

One question I have:

Day 9 — Cross-cultural Calendars

What I learned:

One question I have:

Day 10 — Synthesis and Quiz

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — The Moon Story

Module: Moon Phases & Tithi (Primary) · **Time:** 30 min

Goal

Children meet the Moon as a ball that the Sun lights up, and understand at a basic level why it changes shape.

Materials

- A torch (flashlight)
- A small white ball (a table-tennis ball or styrofoam ball)
- The room dark enough that the torch beam is visible

The flow

Settle + question (3 min)

Sit on the mat. Teacher: *“Today we are going to meet the Moon.”*

Quick question round: *“Did you see the Moon last night? What shape was it?”* Take 4–5 answers, sketch each one on the board.

The story (8 min)

The Moon is a ball.

Imagine a giant white ball, very far away in the sky. That ball is the Moon. It does not make its own light — it is a dark, rocky ball.

Then the Sun shines on it. The Sun is so bright that one half of the Moon lights up. (Hold up the white ball; shine the torch on one side.) Look — half of this ball is bright, and half is dark.

Now — the Moon doesn’t stay still. It travels around the Earth — that’s us. (Walk slowly around a stationary child, holding the ball at arm’s length.)

As the Moon travels, sometimes we see all of the lit side, sometimes only half of the lit side, sometimes none of the lit side. That’s why the Moon looks different every night.

When we see ALL of the lit side, that’s a *pūrṇimā* — a full moon.

When we see NONE of the lit side, that’s an *amāvasyā* — a new moon. The Moon is still there — we just can’t see it.

The demo (12 min)

Set up: - One child stands in the centre = Earth. - Another child holds the torch = Sun. - The teacher walks the ball around Earth in 4 stopping points: 1. Ball between Sun and Earth → Earth sees the dark side (*amāvasyā*). 2. Ball 90° away → Earth sees half lit (half Moon). 3. Ball opposite the Sun → Earth sees full lit (*pūrṇimā*). 4. Ball 90° away on the other side → Earth sees half lit (other half Moon).

At each stop, ask the Earth-child: “*What do you see?*” The class chorus-answers.

Repeat with a few other children rotating roles. Spend the full 12 minutes — children love this and need the repetition.

Word practice (4 min)

Chant 3 times together (clap on each syllable):

am - mah - vāh - yah — the night with no Moon *poor - ni - mah* — the night with the full Moon

Wrap-up (3 min)

- Each child draws one Moon shape on paper — whichever shape they remember best from the demo.
- Hold up 3 drawings.
- **Homework:** *Look at the Moon tonight. Whatever shape you see, draw it in your workbook. If it’s cloudy, ask a grown-up to look it up on a phone.*

Homework / take-home

- **Start your Moon Diary tonight.** Look up. Draw what you see. Note: clear or cloudy.

Differentiation

- **Tier up:** Draw two phases (last night’s + tonight’s predicted).
- **Tier down:** Just remember the word *pūrṇimā* (the easier one). *Amāvasyā* can come later this week.

Teacher’s quick notes

- The single most important sentence today: “**The Sun lights the Moon. The Moon doesn’t make its own light.**” Repeat it often.
- Some children will need the demo repeated. That’s fine — that’s the whole class.
- Don’t worry about Sanskrit spelling. They’ll get it through repetition over the week.

Day 2 — The Two Halves

Module: Moon Phases & Tithi (Primary) · **Time:** 30 min

Goal

Children learn that the month has two halves — one where the Moon grows, one where it shrinks — and that *amāvasyā* and *pūrṇimā* are the special anchor nights.

Materials

- The torch and ball from Day 1
- A long strip of paper or a chalk line on the floor
- Drawing paper + crayons

The flow

Settle + recall (4 min)

Sit on the mat. Teacher: “*Did anyone see the Moon last night?*” Take 3–4 answers. “*Did you draw it in your diary?*” Show one or two diary drawings.

Chant once together: “*am - mah - vahs - yah! poor - ni - mah!*”

Story (6 min)

The Moon takes a journey.

The Moon goes around the Earth — but it takes a long time. About 30 days.

Halfway through the journey, the Moon is on the OPPOSITE side from the Sun. That’s when we see the WHOLE bright side. That’s *pūrṇimā*.

At the start of the journey — and again at the end — the Moon is on the SAME side as the Sun. We can’t see the bright side at all. That’s *amāvasyā*.

So between *amāvasyā* and *pūrṇimā* — the FIRST half of the journey — the Moon gets bigger and bigger every night. That’s the **growing half**.

Between *pūrṇimā* and *amāvasyā* — the SECOND half of the journey — the Moon gets smaller and smaller. That’s the **shrinking half**.

Demo + walk (10 min)

Draw a long line on the floor (or on the board). At the left end write *amāvasyā* and draw a dark circle. At the right end write *pūrṇimā* and draw a bright full circle.

In the middle, draw 5 stages between them — a thin sliver, a half-moon, a 3/4 moon, etc.

Now do the same for the second half — *pūrṇimā* shrinks back through 3/4, half, sliver, to *amāvasyā*.

Have 5 children walk along the line, holding a paper Moon shape at each station. The rest of the class chants “GROWING” while they walk left-to-right and “SHRINKING” while they walk right-to-left.

Drawing (7 min)

Each child draws TWO Moons on paper: 1. A *pūrṇimā* — a full bright circle. Label it “pūrṇimā.” 2. An *amāvasyā* — an empty circle (just the outline, dark inside, or a black filled circle with a tiny sliver). Label it “amāvasyā.”

Wrap-up (3 min)

- Hold up 2–3 drawings.
- Chant once more: “*am - mah - vāh - yah! poor - ni - mah!*”
- **Homework:** Look at the Moon tonight. Is the Moon growing or shrinking? (Ask a grown-up if you don’t know.)

Homework

- Continue the Moon Diary. Add “growing” or “shrinking” next to tonight’s sketch.

Differentiation

- **Tier up:** Draw all the Moon shapes on the line — not just *pūrṇimā* and *amāvasyā*.
- **Tier down:** Just draw *pūrṇimā* today. *Amāvasyā* tomorrow.

Teacher’s quick notes

- Some children will say “the Moon is hiding” on *amāvasyā*. That’s a lovely phrasing — agree, then add: “and the Sun’s bright light is making it hard for us to see anyway.”
- Watch for the misconception “the Earth’s shadow makes the new moon.” It doesn’t (that’s eclipses). Today we don’t unpack that — just say “no, it’s because we can’t see the bright side.”
- The words “growing” and “shrinking” are easier than *śukla pakṣa* / *kṛṣṇa pakṣa*. Stick with English at Primary.

Day 3 — The Moon’s House Visits

Module: Moon Phases & Tithi (Primary) · **Time:** 30 min

Goal

Children meet the idea that the Moon visits a different “house” (star pattern) every night — a folktale-style framing of the 27 *nakṣatras*. The Sanskrit word *nakṣatra* is introduced softly without drilling.

Materials

- A printed Moon and 27 paper “house” stars arranged in a circle on the floor (or on a wall poster)
- Drawing paper

The flow

Settle + recall (3 min)

Chant: “*am - mah - vāhs - yah! poor - ni - mah!*” — three times.

Recall: “*Is the Moon growing or shrinking right now?*” (Refer to the day’s actual sky if children have observed.)

Story (12 min)

The Moon Visits Its House Family

Long, long ago, the Moon — whose name is *Soma* — had 27 sisters. Each sister lived in her own house, high up in the sky. Together they were the family of *nakṣatras* — the “star houses.”

Every night, the Moon visited ONE of his 27 sister houses. He never visited two on the same night. So one night he was in the house called *Aśvinī*. The next night, *Bharanī*. The next, *Kṛttikā*. And so on, all 27 sisters, one per night.

After 27 nights, he had visited every sister. Then he started over — back to *Aśvinī*.

(Children love this. Walk a paper Moon around the 27 stars on the floor, naming a few of them as you go.)

Now — the Moon also changes shape every night, as we learned. Sometimes he is full (*pūrṇimā*), sometimes he disappears (*amāvasyā*). Some old stories say his sisters are happy when he is full, and worried when he disappears.

But every month, he comes back to the same sister he visited a month ago. So if tonight the Moon is in the house of *Citrā*, in about a month, the Moon will visit *Citrā* again.

That’s why our months in India are named after the sisters. The month *Caitra* is named after *Citrā*. The month *Kārtika* is named after *Kṛttikā*. The Moon’s sisters give the months their names.

Activity (10 min)

Each child: 1. Draws the Moon in the middle of a paper. 2. Draws 4 little “house stars” around the Moon (we’ll do 4, not 27). 3. Labels one of the houses with the name *Aśvinī* (or any *nakṣatra* name the teacher chooses). 4. Draws arrows showing the Moon’s path from one house to the next.

This is a drawing exercise. Don’t drill the names.

Wrap-up (5 min)

- Show a few drawings.
- “*What did we learn? The Moon visits a new house every night. There are 27 houses in all. The Moon’s sisters give our months their names.*”
- **Homework:** *Look at the Moon tonight. What shape? Write the date.*

Homework

- Continue the Moon Diary.

Differentiation

- **Tier up:** Try drawing all 27 houses in a ring around the Moon.
- **Tier down:** Just draw the Moon and one house. The story is the lesson — the drawing supports it.

Teacher’s quick notes

- The “27 sisters” framing is a teaching tool drawing on traditional stories (the Dakṣa’s daughters narrative appears in several *Purāṇas*, with the Moon as *Soma* / *Candra*). At Primary band, frame it as “an old story” — don’t claim a single canonical source.
- The point: children learn that the Moon’s position in the sky changes night by night, in a 27-night cycle (the *sidereal* month — different from the 30-day *synodic* month, but we don’t need that distinction yet).
- If a child asks “why don’t the houses come back to the same place every 30 nights?” — that’s a great question, and you can tell them they’ll learn about it in Middle school. (The answer: because Earth has moved around the Sun during the 27 days too.)
- Some children may know specific nakṣatras from their birth chart. That’s fine — let them share, but don’t dwell on astrology.

Day 4 — Building a Tithi Calendar

Module: Moon Phases and Tithi (Primary) · **Time:** 30 min

Goal

Construct a tithi calendar for the current month using observation and computation.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

Today’s idea (10 min)

- *kṛṣṇa pakṣa* (IAST: kṛṣṇa pakṣa; lit. “waning fortnight (full → new)”)

- Let's make a moon calendar for this month!

Show a printed month calendar. Together, look up today's tithi. Mark it with a moon-sticker.

Each child gets a worksheet with 28 boxes (one per day). They draw the moon shape they expect to see each evening, going from now to a month from now. Tomorrow we'll check and correct.

Activity (8 min)

A simplified version of *10-Day Moon Observation Diary* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 5 — Lunar Months and Festivals

Module: Moon Phases and Tithi (Primary) · **Time:** 30 min

Goal

Map the 12 lunar months (caitra, vaiśākha, ...) to major festivals.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: "What's one thing you remember from yesterday?"

Today's idea (10 min)

- *tithi* (IAST: tithi; lit. “lunar day (1/30 of synodic month)”)
- The year has 12 moon-months, each with its own name and feeling:
 - *Caitra* (spring)
 - *Śrāvaṇa* (monsoon)
 - *Kārtika* (Diwali month)
 - *Phālguna* (Holi month) — and many more!

Show pictures of 4 festivals. Ask: “Which moon-month is this festival in?” Children sort cards.

Sing the months: “*Caitra, Vaiśākha, Jyaiṣṭha, Āṣāḍha...*” (Through all 12.)

Activity (8 min)

A simplified version of *10-Day Moon Observation Diary* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 6 — Panchanga — Five Limbs

Module: Moon Phases and Tithi (Primary) · **Time:** 30 min

Goal

Identify the five components: tithi, vāra, nakṣatra, yoga, karaṇa.

Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *amāvasyā* (IAST: amāvasyā; lit. “new moon”)
- The pañcāṅga has 5 parts — like 5 fingers on a hand!
 1. *tithi* — moon-day
 2. *vāra* — weekday (Mon, Tue, ...)
 3. *nakṣatra* — star-house the moon is visiting
 4. *yoga* — Sun-Moon meeting
 5. *karaṇa* — half-moon-day

Show today's pañcāṅga on the board. Children copy the 5 names into their workbook with one word each.

Activity (8 min)

A simplified version of *10-Day Moon Observation Diary* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

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- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 7 — Why Eclipses Happen

Module: Moon Phases and Tithi (Primary) · **Time:** 30 min

Goal

Explain solar/lunar eclipses using orbital plane geometry (5° tilt of Moon's orbit).

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *pūrṇimā* (IAST: pūrṇimā; lit. “full moon”)
- Sometimes the Moon hides the Sun — that's a *solar eclipse*!
- Sometimes the Earth's shadow falls on the Moon — that's a *lunar eclipse*!

Use the torch + ball + child's head from earlier days. Show: - Ball blocks light → solar eclipse (child's face goes dark). - Child's head blocks light → lunar eclipse (ball goes dark).

SAFETY RULE: Never look at the Sun with eyes! Children repeat 3 times.

Activity (8 min)

A simplified version of *10-Day Moon Observation Diary* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 8 — Modern Lunar Science

Module: Moon Phases and Tithi (Primary) · **Time:** 30 min

Goal

Phases as Sun's illumination of Moon; mention libration, tides.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

Today's idea (10 min)

- *pañcāṅga* (IAST: pañcāṅga; lit. “five-limbed Indian calendar (tithi, vāra, nakṣatra, yoga, karaṇa)”)
- The Moon isn't really changing shape — the same ball is always there!
- The Sun lights up half of it. We see *different parts of the lit half* as the Moon goes around Earth.

Demo: hold a half-painted ball (one side bright, one side dark). Walk it around a child. The child describes what they see from each angle.

“*The Moon is always whole. We only see the bright part.*”

Activity (8 min)

A simplified version of *10-Day Moon Observation Diary* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 9 — Cross-cultural Calendars

Module: Moon Phases and Tithi (Primary) · **Time:** 30 min

Goal

Compare with Hijri (Islamic), Chinese, and Hebrew lunar calendars.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: *"What's one thing you remember from yesterday?"*

Today's idea (10 min)

- *candra* (IAST: candra; lit. "Moon")
- People all over the world watch the Moon, but each tradition counts months a bit differently.
- Some use only Moon (Islamic Hijri). Some use Sun + Moon (Indian, Chinese, Hebrew). Some use only Sun (modern Gregorian — like the school year).

Show 4 calendar pictures and chant: *"India and China — Moon plus Sun! Islam — Moon alone! Modern world — Sun alone!"*

Each child colours a small "moon flag" with the calendar they think they use most.

Activity (8 min)

A simplified version of *10-Day Moon Observation Diary* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See homework.md.)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Day 10 — Synthesis and Quiz

Module: Moon Phases and Tithi (Primary) · **Time:** 30 min

Goal

Final quiz and 10-day moon-diary share.

Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

The flow

Settle + chant (3 min)

Daily chant or breathing exercise.

Story recall (4 min)

Briefly recall yesterday's lesson by asking: *“What's one thing you remember from yesterday?”*

Today's idea (10 min)

- *pakṣa* (IAST: pakṣa; lit. “fortnight (half-month)”)
- Today we celebrate everything we learned about the Moon!

Each child shares ONE thing from their moon diary. *“Did the Moon look how you thought it would?”*

Make a class moon-mural: a circle with all the children's moon drawings stuck around the outside. Sing the tithi rhyme one more time.

Take the quiz with stickers (not stress) — every child gets a “Moon Watcher” sticker.

Activity (8 min)

A simplified version of *10-Day Moon Observation Diary* (see [activities/](#)). Draw, sing, or act it out.

Wrap-up (3 min)

Each child shares one word that describes today.

Homework

(See [homework.md](#).)

Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

Homework

Homework — Moon Phases and Tithi (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — The Moon and Its Shapes** · Draw about today's concept. (~5 min)
2. **Day 2 — Two Fortnights — Pakṣa** · Draw about today's concept. (~5 min)
3. **Day 3 — Fifteen Tithis per Pakṣa** · Draw about today's concept. (~5 min)
4. **Day 4 — Building a Tithi Calendar** · Draw about today's concept. (~5 min)
5. **Day 5 — Lunar Months and Festivals** · Draw about today's concept. (~5 min)
6. **Day 6 — Panchanga — Five Limbs** · Draw about today's concept. (~5 min)
7. **Day 7 — Why Eclipses Happen** · Draw about today's concept. (~5 min)
8. **Day 8 — Modern Lunar Science** · Draw about today's concept. (~5 min)
9. **Day 9 — Cross-cultural Calendars** · Draw about today's concept. (~5 min)
10. **Day 10 — Synthesis and Quiz** · Draw about today's concept. (~5 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.